

Experiment NO:3 :- Standardization of a borax solution provided using standard HCl

Theory

A solution of a borax is titrated against standard HCl. The reaction between borax and HCl is,



The stoichiometric ratio of the reaction is $\text{Na}_2\text{B}_4\text{O}_7 : \text{HCl} = 1:2$. Thus if an accurately measured volume of the standard HCl solution is titrated against the unknown borax solution, the concentration of borax in the solution may be determined with accuracy. The indicator methyl orange is used to indicate the end point of the reaction.

Procedure

Pipette out 25.00 cm^3 of the borax solution in to a titration flask. Add a drop of methyl orange indicator in to the titration flask, and titrate the solution against the standard HCl solution in the burette until the color change from yellow to pink. Repeat the titration until two conservative measurements are obtained.

Note: Before the equivalence point, the solution is very slightly acidic due to the weak boric acid that is formed by the reaction. This has no influence on methyl orange which has a working pH range 3.1 – 4.

Experiment NO:4 :- Standardization of NaOH solution provided using standard Borax solution

Theory

Borax when treated with equivalent amount of HCl, gives the weak acid H_3BO_3 (boric acid) as stated earlier.



H_3BO_3 is a strong monoprotic acid in the presence of certain organic polyhydroxides. The stoichiometry of the reaction between NaOH and the boric acid polyhydroxide complex.

Procedure

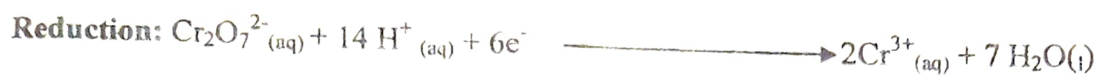
Pipette out 25.00 cm^3 of the borax solution in to a titration flask and add exact equivalent amount of HCl from experiment 2 to neutralize this borax. Add approximately 1g of manitol and 2 drop of phenolphthalein indicator in to the titration flask, and titrate the solution against NaOH solution in the burette until the color change from colorless to red. Repeat the titration until two conservative measurements are obtained.

Calculate the concentration of NaOH solution using titration data

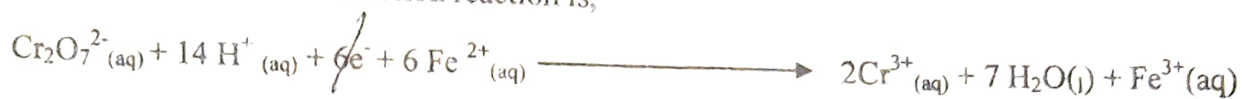
Experiment NO:5 :- Standardization of $K_2Cr_2O_7$ using standard $Fe(NH_4)(SO_4)_2$ solution

Theory

A solution of $K_2Cr_2O_7$ is standardized against standard solution. The reaction taking place are:



Therefore, the **overall reduction** reaction is,



The stoichiometric ratio of the reaction is $Cr_2O_7^{2-} : Fe^{2+} = 1 : 6$. Thus, if an accurately measured aliquot of a certain volume of a solution of known Fe^{2+} concentration is titrated against the unknown solution, the concentration of the unknown $Cr_2O_7^{2-}$ solution may be determined with accuracy. An indicator known as diphenylamine is used to indicate the end of the reaction.

Procedure:

Pipette out 25.00 cm^3 of the standard ferrous ammonium sulphate solution ($M \text{ mol/dm}^3$, say) in to a titration flask. Add 15 cm^3 of dill H_2SO_4 , 3 cm^3 of 85% H_3PO_4 and 2 drops of diphenylamine indicator to the titration flask, and titrate the solution against the $K_2Cr_2O_7$ solution in the burette until the color changes from green to dark blue. Repeat the titration until two consecutive measurements are obtained.

Notes:

1. Diphenylamine has a redox potential intermediate between those of Fe^{2+}/Fe^{3+} and $Cr_2O_7^{2-}/Cr^{3+}$. Its reduced form is colorless. As long as Fe^{2+} is present in the solution, it remains in its reduced form. After all the Fe^{2+} has been consumed, the next drop of added oxidizes the indicator. The oxidized form is dark blue color.
2. The H_3PO_4 is added to remove Fe^{3+} ions that are formed in the reaction. Fe^{3+} ions form a ferric phosphate complex which does not dissociate, thereby removing free Fe^{3+} ions from solution. If not removed, these ions can oxidize the diphenylamine indicator.
3. H_3PO_4 also lowers the redox potential of the Fe^{2+}/Fe^{3+} system, thereby making the equivalence point potential closer to that of the indicator.

In the sixth experiment, the standardized $K_2Cr_2O_7$ solution from the fourth experiment will be used to determine the concentration of a $Na_2S_2O_3$ solution of unknown strength.